

PCA in R

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PCA in R

Import Library

```
library(factoextra)
```

```
## Loading required package: ggplot2
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
library(stats)  
library(cluster)
```

```
## Warning: package 'cluster' was built under R version 4.3.3
```

```
library(factoextra)  
library(ggplot2)  
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.3.3
```

```
## Loading required package: lattice
```

```
## Warning: package 'lattice' was built under R version 4.3.3
```

```
library(ggbiplot)
```

Step 1: Scale

```
data_scaled <- scale(data)
```

PCA is affected by the scale of the variables, so standardization is usually necessary.

Step 2: apply prcomp()

```
pca_data <- prcomp(
  data_scaled,
  center=TRUE,
  scale.=TRUE # if you didn't scale the data
)
```

If scaled

```
pca_data <- prcomp(
  data_scaled,
  center=FALSE,
  scale.=FALSE # if you didn't scale the data
)
```

Summary

```
summary(pca_data)
```

Output includes:

Statistic	Description
Standard deviation	The square roots of the eigenvalues of the covariance/correlation matrix. It shows how much variance each principal component (PC) captures in absolute terms.
Proportion of Variance	The percentage of the total variance in the dataset explained by each PC.
Cumulative Proportion	The cumulative percentage of total variance explained as you add more PCs. Useful for deciding how many components to keep.

What you can access in the `pca_data` object:

Component	Description
<code>\$sdev</code>	Standard deviations of the principal components (used to compute variance).
<code>\$rotation</code>	The matrix of eigenvectors (i.e., loadings). Columns = PCs, rows = original features. Shows how each variable contributes to each PC.
<code>\$center</code>	The mean of each variable used for centering.
<code>\$scale</code>	The standard deviation used for scaling (if <code>scale. = TRUE</code>).
<code>\$x</code>	The principal component scores , i.e., the projected data in the new PC space. Each row = observation, each column = PC.

Step 3: Scree Plot

We need to use Scree Plot to see and help to decide how many PCs to keep.

```
fviz_eig(pca_data)
```

How to Interpret It:

1. Look for the “elbow”

- This is the point where the slope of the curve **levels off**.
- It indicates that **adding more PCs beyond this point contributes little extra variance**.
- **Keep PCs before the elbow**, discard those after.

2. Variance Explained

- Each point shows **how much variance a PC explains**.
- The **higher the point**, the more useful that PC is for summarizing the data.

Rule of Thumb:

- If the **first few PCs explain 70–90% of the variance**, you may choose to keep them.

Biplot

```
ggbiplot(pca_data)
```

What does a biplot show?

1. Points (Observations)

- Each point represents a **data sample** (an observation in your dataset).
- Their positions are based on their scores on the selected principal components (usually PC1 and PC2).

2. Arrows (Variables)

- Each arrow represents a **variable (feature)** from your original data.
- The **direction** and **length** of the arrow show how strongly that variable contributes to the PCs.

Element	Interpretation
Length of arrows	Variables with longer arrows are more strongly correlated with the PCs shown.
Angle between arrows	Small angle -> high positive correlation. 90° -> no correlation. 180° -> negative correlation.

Element	Interpretation
Direction of arrows	Shows which way the variable increases. Observations in that direction have higher values for that variable.
Position of points	Samples that are close together are similar in terms of the principal components.

Example

Our goal is to predict the iris Species based on the other 4 attributes given. Subsequently we wish to compare our data-driven clustering to that of the true species classification. For each of the following six clustering methods

- K-mean;
- Hierarchical: Ward, Single-linkage, Complete-linkage, Average-linkage, Centroid

K-means

Data-Preprocessing

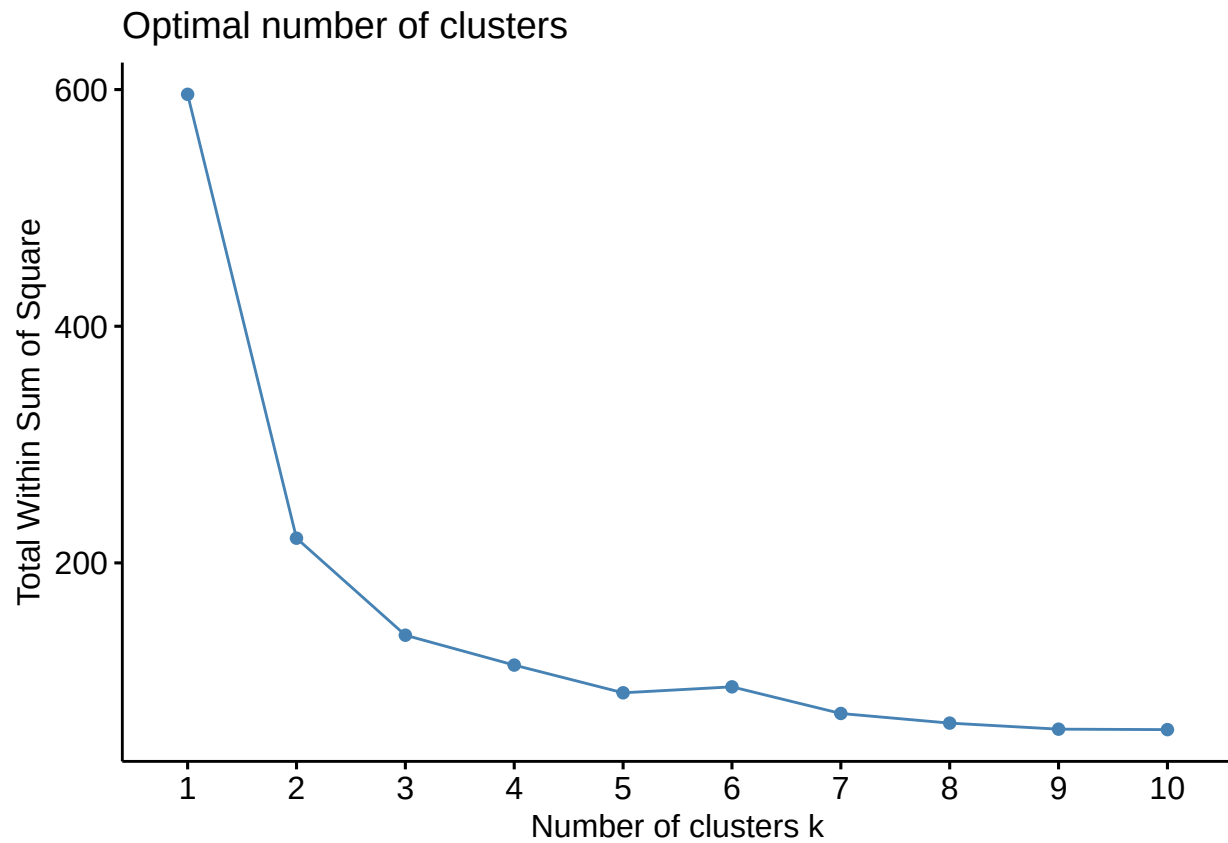
```
# omit NAs
data <- na.omit(iris)

# scale data
scaled_data <- scale(data[, -5]) #don't scale the response
```

Optimal number of clusters

```
# K-means
kmeans_fit <- kmeans(
  scaled_data,
  centers = 3,
  nstart = 25
)

# Scree-plot
fviz_nbclust(
  scaled_data,
  kmeans,
  method="wss"
)
```

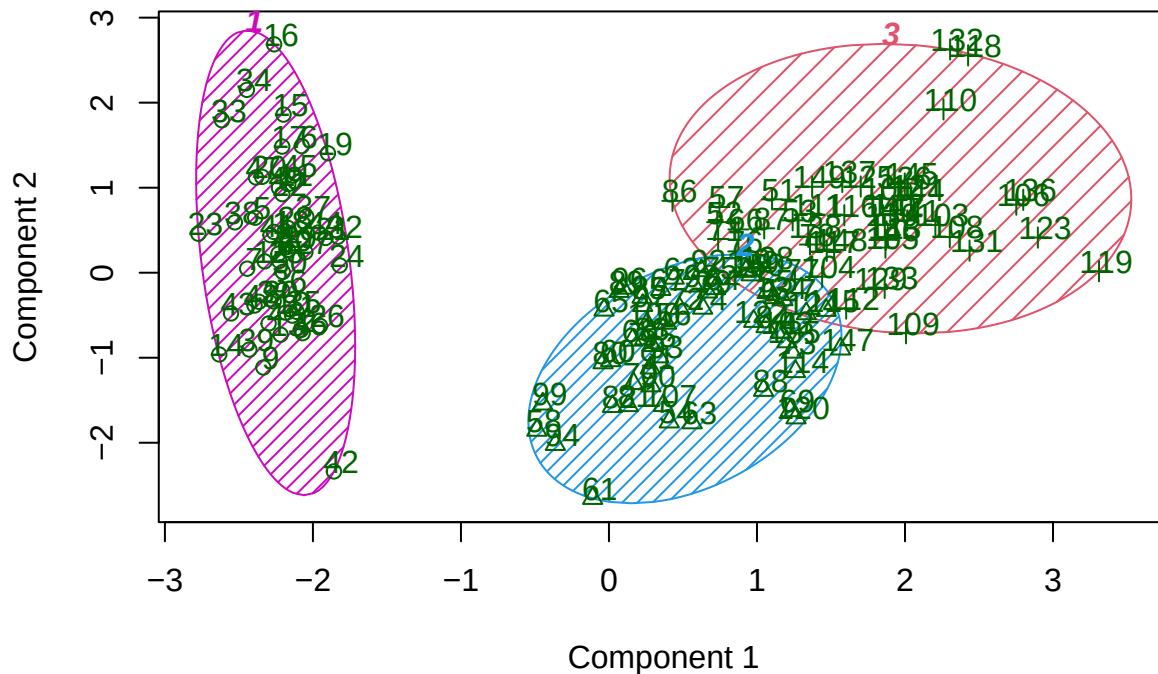


3 clusters seems good

2D Cluster Plot

```
clusplot(  
  scaled_data,  
  kmeans_fit$cluster,  
  main = "K-means Clustering",  
  color = TRUE,  
  shade = TRUE,  
  labels = 2,  
  lines = 0  
)
```

K-means Clustering



These two components explain 95.81 % of the point variability.

Confusion Matrix

```
table(
  kmeans_fit$cluster,
  data$Species
)
```

```
##
##      setosa versicolor virginica
##  1      50           0           0
##  2       0          39          14
##  3       0          11          36
```

```
#mapping
clusters <- factor(
  kmeans_fit$cluster,
  levels = c(3,1,2),
  labels = c("setosa","versicolor","virginica")
)

confusionMatrix(
  clusters,
  data$Species
)
```

```
## Confusion Matrix and Statistics
```

```
##
##           Reference
## Prediction  setosa versicolor virginica
##   setosa      0         11         36
##   versicolor  50         0         0
##   virginica   0         39         14
##
## Overall Statistics
##
##           Accuracy : 0.0933
##           95% CI : (0.052, 0.1516)
##   No Information Rate : 0.3333
##   P-Value [Acc > NIR] : 1
##
##           Kappa : -0.36
##
## McNemar's Test P-Value : <2e-16
##
## Statistics by Class:
##
##           Class: setosa Class: versicolor Class: virginica
## Sensitivity           0.0000           0.0000           0.28000
## Specificity           0.5300           0.5000           0.61000
## Pos Pred Value        0.0000           0.0000           0.26415
## Neg Pred Value        0.5146           0.5000           0.62887
## Prevalence            0.3333           0.3333           0.33333
## Detection Rate        0.0000           0.0000           0.09333
## Detection Prevalence  0.3133           0.3333           0.35333
## Balanced Accuracy     0.2650           0.2500           0.44500
```

so the confusion matrix is

```
50 0 0
0 39 14
0 11 36
```

So the accuracy is

```
(39+50+36)/150
```

```
## [1] 0.8333333
```

Hierarchical Clustering

Distance Matrix

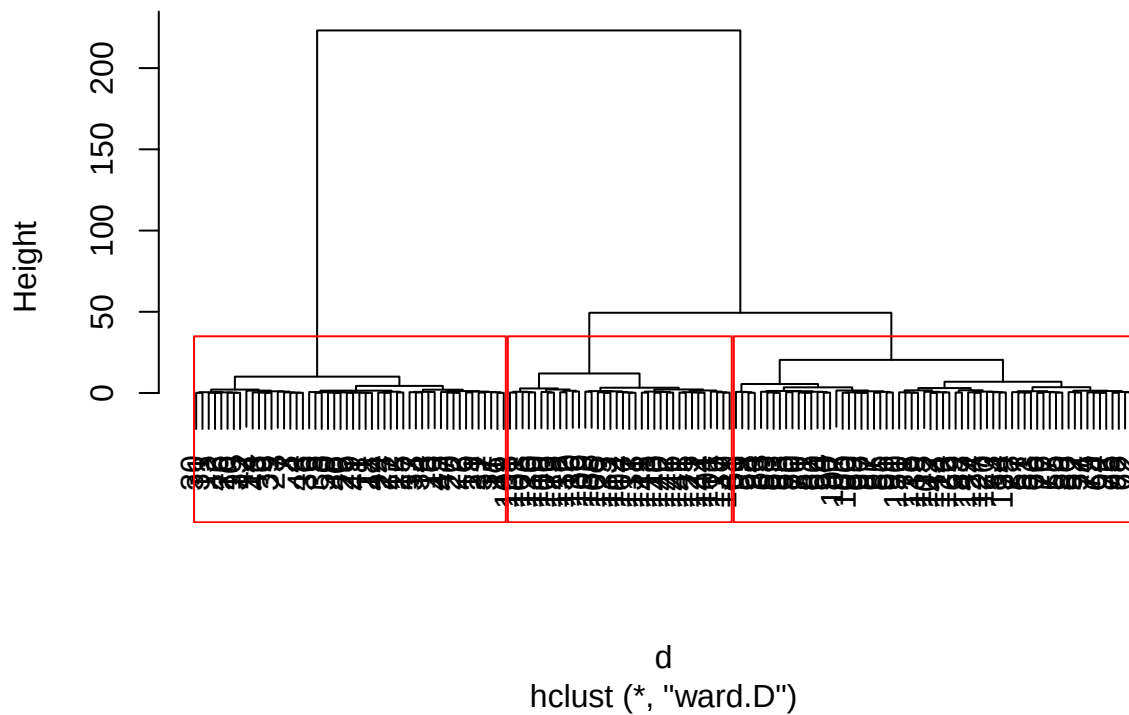
```
d <- dist(
  data,
  method = "euclidean"
)
```

```
## Warning in dist(data, method = "euclidean"): NAs introduced by coercion
```

Ward.D

```
# fit
H.fit <- hclust(
  d,
  method = "ward.D"
)
# plot
plot(H.fit)
# Add rectangles for k clusters
rect.hclust(
  H.fit,
  k=3,
  border = "red"
)
```

Cluster Dendrogram



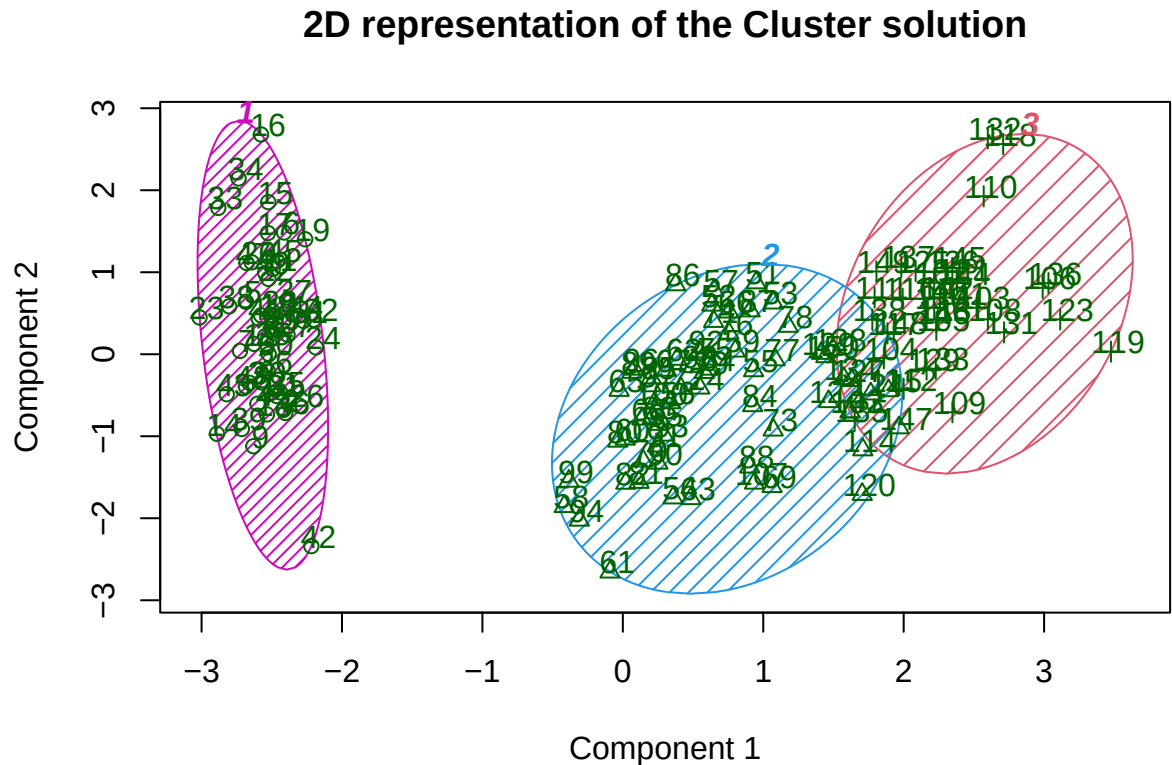
```
groups <- cutree(
  H.fit,
  k=3
)

clusplot(
  data,
  groups,
```

```

main = "2D representation of the Cluster solution",
color = TRUE,
shade = TRUE,
labels = 2,
lines = 0
)

```



2D Cluster Plot

These two components explain 95.02 % of the point variability.

```

table(
  data$Species,
  groups
)

```

Confusion Matrix

```

##           groups
##           1  2  3
##  setosa    50  0  0
##  versicolor 0 50  0
##  virginica  0 14 36

```

```

clusters <- factor(
  groups,
  levels =1:3,

```

```

labels = c("setosa","versicolor","virginica")
)

confusionMatrix(
  data$Species,
  clusters
)

```

```

## Confusion Matrix and Statistics
##
##              Reference
## Prediction  setosa versicolor virginica
##   setosa      50         0         0
##   versicolor  0         50         0
##   virginica   0         14        36
##
## Overall Statistics
##
##              Accuracy : 0.9067
##              95% CI : (0.8484, 0.948)
##   No Information Rate : 0.4267
##   P-Value [Acc > NIR] : < 2.2e-16
##
##              Kappa : 0.86
##
##   McNemar's Test P-Value : NA
##
## Statistics by Class:
##
##              Class: setosa Class: versicolor Class: virginica
## Sensitivity              1.0000              0.7812              1.0000
## Specificity              1.0000              1.0000              0.8772
## Pos Pred Value           1.0000              1.0000              0.7200
## Neg Pred Value           1.0000              0.8600              1.0000
## Prevalence               0.3333              0.4267              0.2400
## Detection Rate           0.3333              0.3333              0.2400
## Detection Prevalence     0.3333              0.3333              0.3333
## Balanced Accuracy        1.0000              0.8906              0.9386

```

so the confusion matrix is

50 0 0

0 50 0

0 14 36

So the accuracy is

```
(50+50+36)/150
```

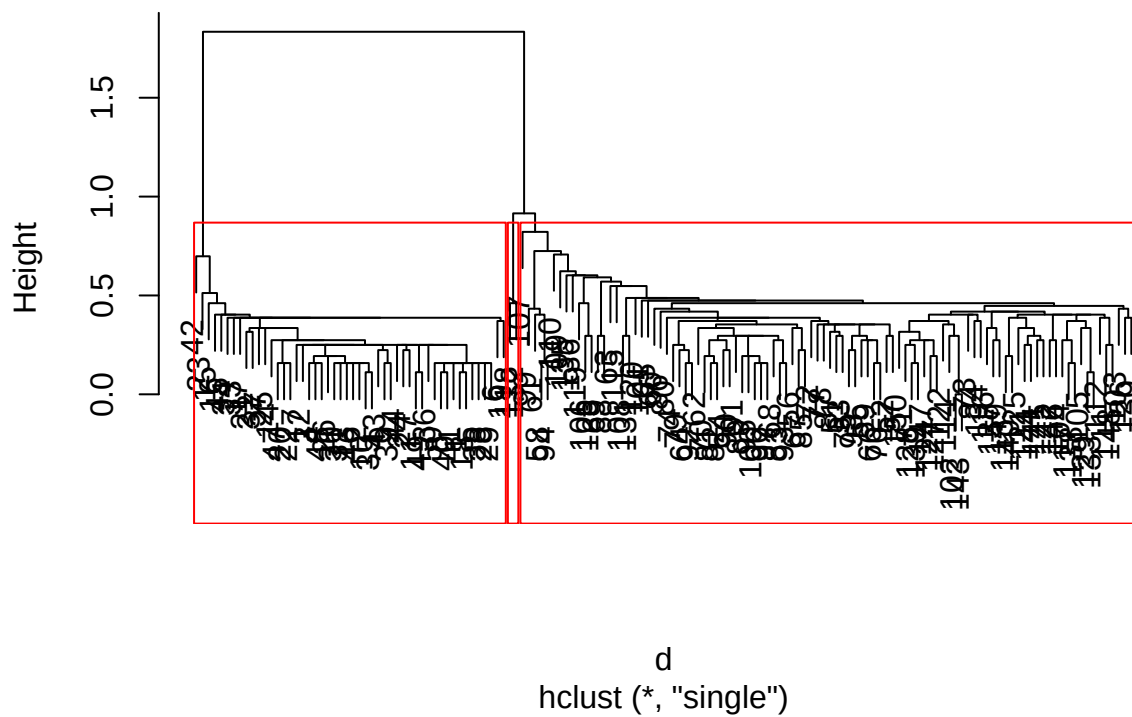
```
## [1] 0.9066667
```

Single linkage

```
# fit
H.fit <- hclust(
  d,
  method = "single"
)
# plot
plot(H.fit)

# Add rectangles for k clusters
rect.hclust(
  H.fit,
  k=3,
  border = "red"
)
```

Cluster Dendrogram

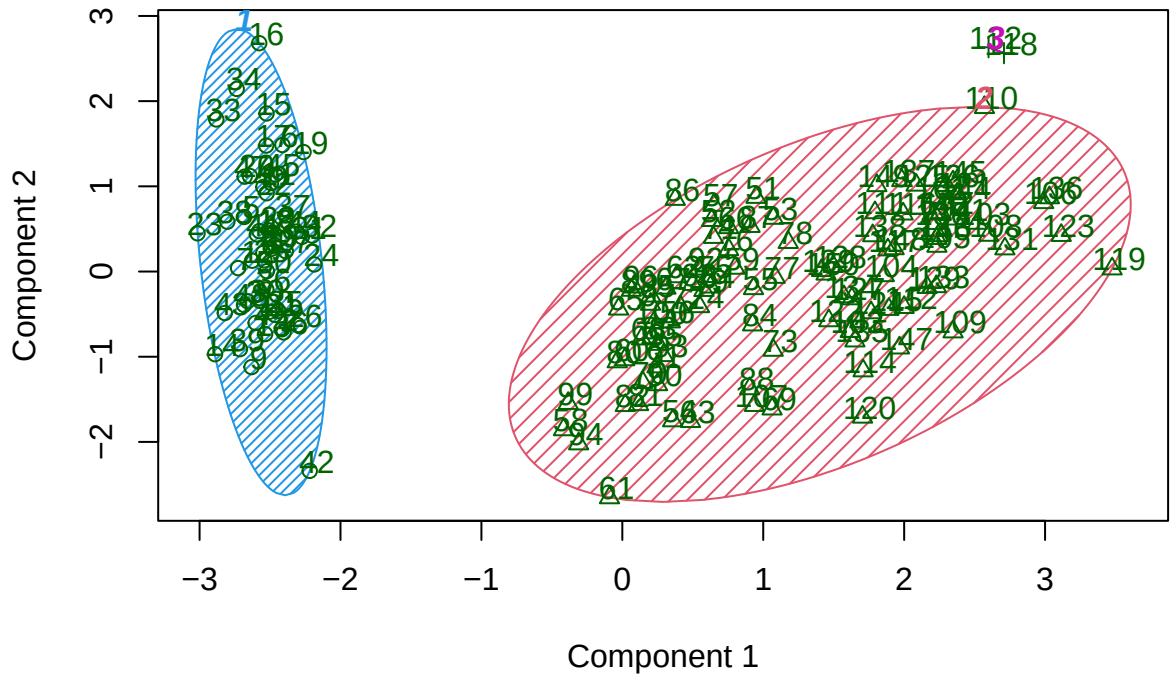


```
groups <- cutree(
  H.fit,
  k=3
)

clusplot(
  data,
```

)

2D representation of the Cluster solution



2D Cluster Plot

These two components explain 95.02 % of the point variability.

```
table(
  data$Species,
  groups
)
```

Confusion Matrix

```
##           groups
##           1  2  3
## setosa      50  0  0
## versicolor  0 50  0
## virginica   0 48  2
```

```
clusters <- factor(
  groups,
  levels = 1:3,
```

```

labels = c("setosa","versicolor","virginica")
)

confusionMatrix(
  data$Species,
  clusters
)

```

```

## Confusion Matrix and Statistics
##
##               Reference
## Prediction  setosa versicolor virginica
##   setosa      50         0         0
##   versicolor  0         50         0
##   virginica   0         48         2
##
## Overall Statistics
##
##               Accuracy : 0.68
##               95% CI : (0.599, 0.7537)
##   No Information Rate : 0.6533
##   P-Value [Acc > NIR] : 0.276
##
##               Kappa : 0.52
##
##   McNemar's Test P-Value : NA
##
## Statistics by Class:
##
##               Class: setosa Class: versicolor Class: virginica
## Sensitivity          1.0000          0.5102          1.00000
## Specificity          1.0000          1.0000          0.67568
## Pos Pred Value        1.0000          1.0000          0.04000
## Neg Pred Value        1.0000          0.5200          1.00000
## Prevalence           0.3333          0.6533          0.01333
## Detection Rate        0.3333          0.3333          0.01333
## Detection Prevalence  0.3333          0.3333          0.33333
## Balanced Accuracy     1.0000          0.7551          0.83784

```

so the confusion matrix is

50 0 0

0 50 0

0 48 2

So the accuracy is

```
(50+50+2)/150
```

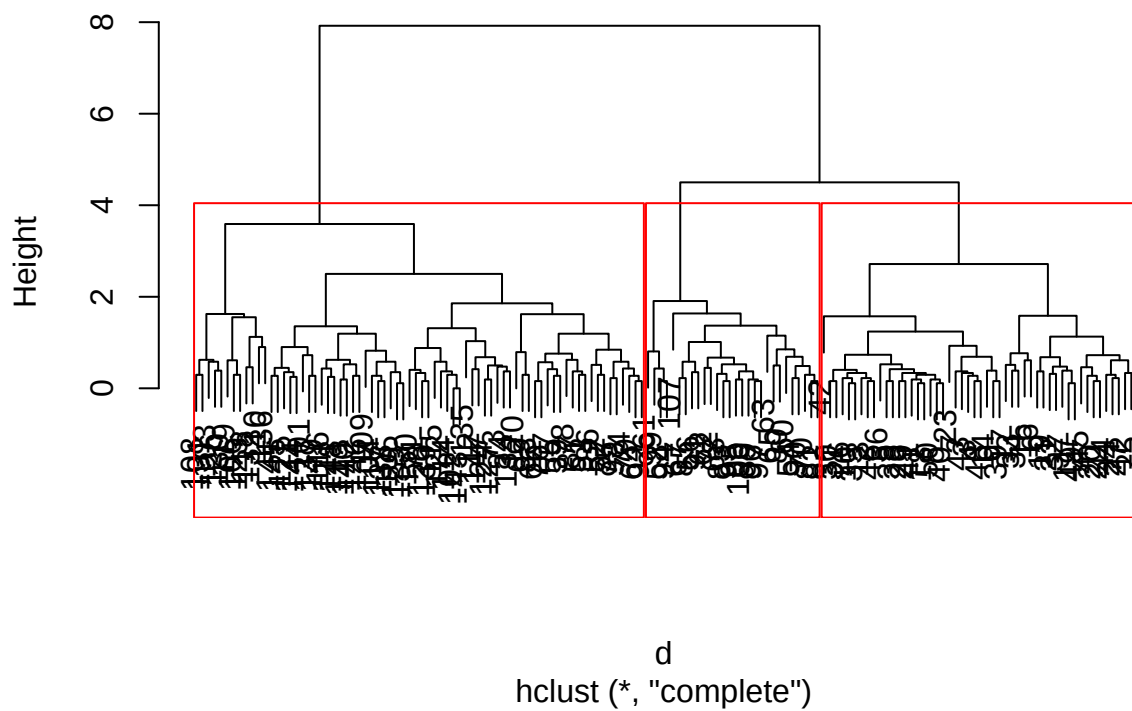
```
## [1] 0.68
```

Complete

```
# fit
H.fit <- hclust(
  d,
  method = "complete"
)
# plot
plot(H.fit)

# Add rectangles for k clusters
rect.hclust(
  H.fit,
  k=3,
  border = "red"
)
```

Cluster Dendrogram



```
groups <- cutree(
  H.fit,
  k=3
)

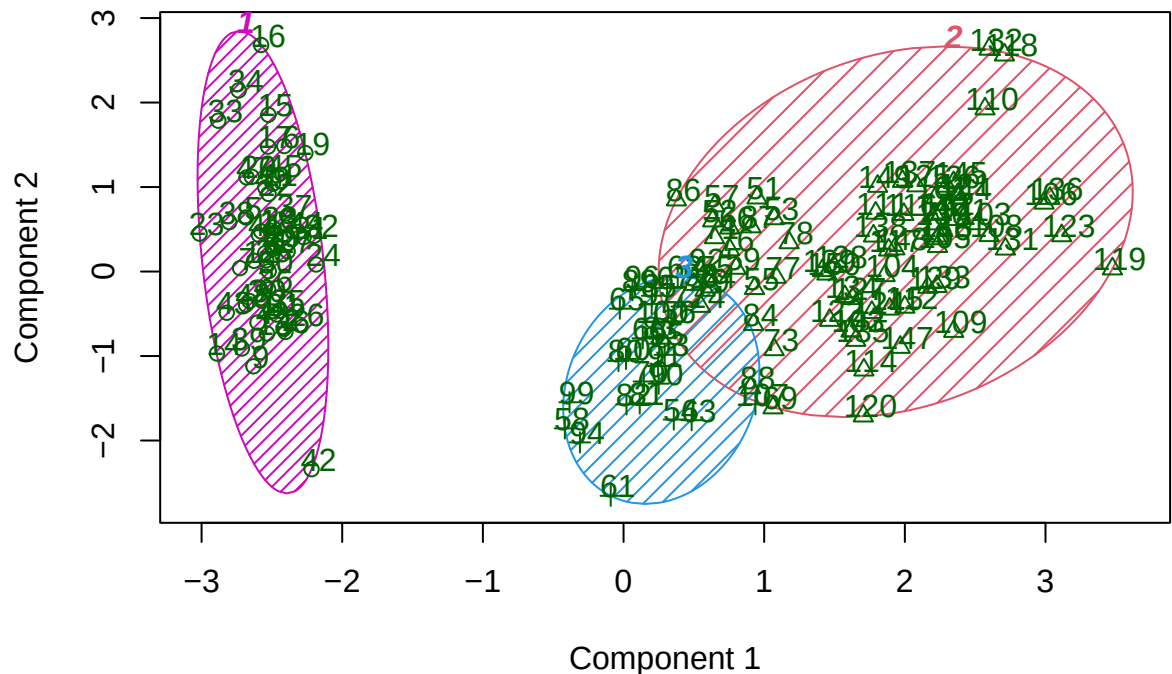
clusplot(
  data,
```

```

groups,
main = "2D representation of the Cluster solution",
color = TRUE,
shade = TRUE,
labels = 2,
lines = 0
)

```

2D representation of the Cluster solution



2D Cluster Plot

These two components explain 95.02 % of the point variability.

```

table(
  data$Species,
  groups
)

```

Confusion Matrix

```

##           groups
##           1  2  3
##  setosa    50  0  0
##  versicolor 0 23 27
##  virginica  0 49  1

```

```

clusters <- factor(
  groups,
  levels =1:3,

```

```

labels = c("setosa","versicolor","virginica")
)

confusionMatrix(
  data$Species,
  clusters
)

```

```

## Confusion Matrix and Statistics
##
##              Reference
## Prediction  setosa versicolor virginica
##   setosa      50         0         0
##   versicolor   0         23        27
##   virginica    0         49         1
##
## Overall Statistics
##
##              Accuracy : 0.4933
##              95% CI : (0.4108, 0.5761)
##   No Information Rate : 0.48
##   P-Value [Acc > NIR] : 0.4028
##
##              Kappa : 0.24
##
##   McNemar's Test P-Value : NA
##
## Statistics by Class:
##
##              Class: setosa Class: versicolor Class: virginica
## Sensitivity              1.0000              0.3194              0.035714
## Specificity              1.0000              0.6538              0.598361
## Pos Pred Value           1.0000              0.4600              0.020000
## Neg Pred Value           1.0000              0.5100              0.730000
## Prevalence               0.3333              0.4800              0.186667
## Detection Rate           0.3333              0.1533              0.006667
## Detection Prevalence     0.3333              0.3333              0.333333
## Balanced Accuracy         1.0000              0.4866              0.317037

```

so the confusion matrix is

50 0 0

0 23 27

0 49 1

So the accuracy is

```
(50+23+1)/150
```

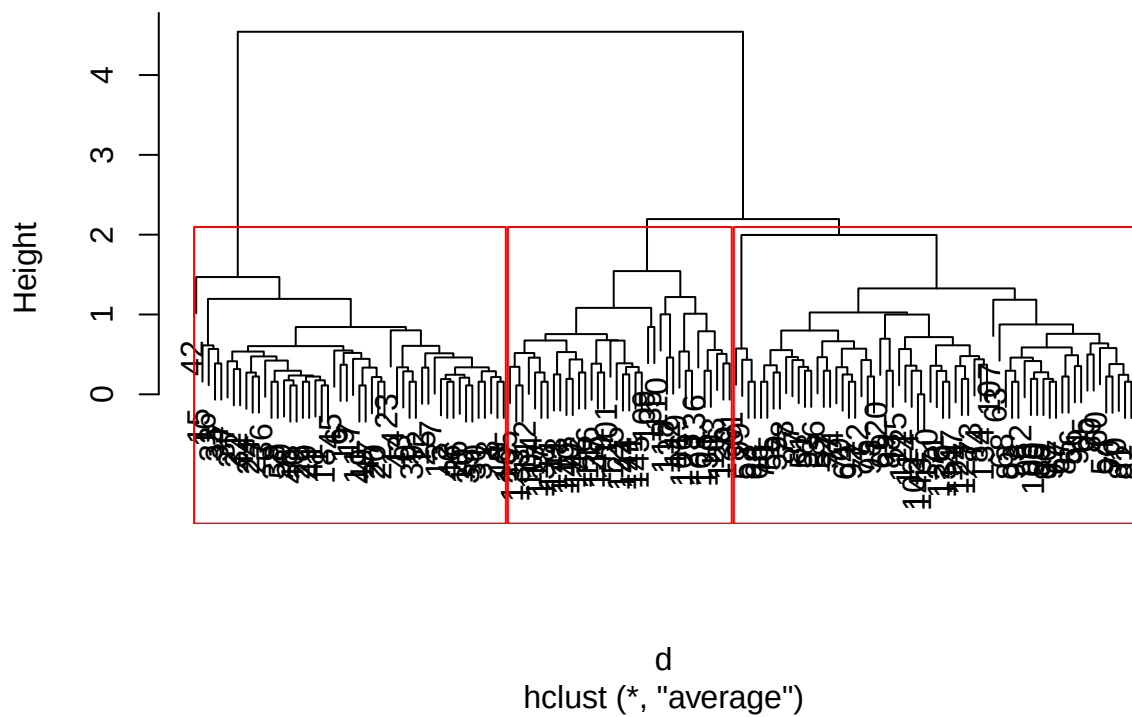
```
## [1] 0.4933333
```

Average

```
# fit
H.fit <- hclust(
  d,
  method = "average"
)
# plot
plot(H.fit)

# Add rectangles for k clusters
rect.hclust(
  H.fit,
  k=3,
  border = "red"
)
```

Cluster Dendrogram



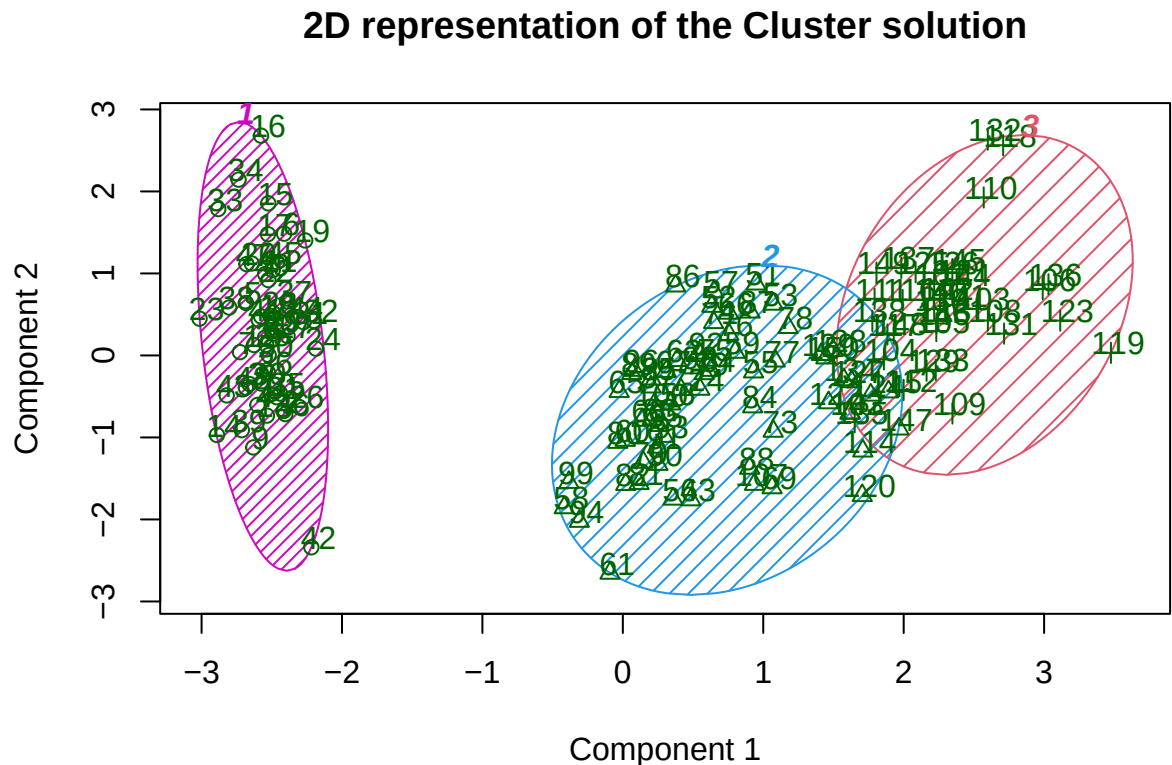
```
groups <- cutree(
  H.fit,
  k=3
)

clusplot(
  data,
```

```

groups,
main = "2D representation of the Cluster solution",
color = TRUE,
shade = TRUE,
labels = 2,
lines = 0
)

```



2D Cluster Plot

These two components explain 95.02 % of the point variability.

```

table(
  data$Species,
  groups
)

```

Confusion Matrix

```

##           groups
##           1  2  3
##  setosa    50  0  0
##  versicolor 0 50  0
##  virginica  0 14 36

```

```

clusters <- factor(
  groups,
  levels =1:3,

```

```

labels = c("setosa","versicolor","virginica")
)

confusionMatrix(
  data$Species,
  clusters
)

```

```

## Confusion Matrix and Statistics
##
##               Reference
## Prediction  setosa versicolor virginica
##   setosa      50         0         0
##   versicolor  0         50         0
##   virginica   0         14        36
##
## Overall Statistics
##
##               Accuracy : 0.9067
##               95% CI : (0.8484, 0.948)
##   No Information Rate : 0.4267
##   P-Value [Acc > NIR] : < 2.2e-16
##
##               Kappa : 0.86
##
##   McNemar's Test P-Value : NA
##
## Statistics by Class:
##
##               Class: setosa Class: versicolor Class: virginica
## Sensitivity           1.0000           0.7812           1.0000
## Specificity           1.0000           1.0000           0.8772
## Pos Pred Value        1.0000           1.0000           0.7200
## Neg Pred Value        1.0000           0.8600           1.0000
## Prevalence            0.3333           0.4267           0.2400
## Detection Rate        0.3333           0.3333           0.2400
## Detection Prevalence  0.3333           0.3333           0.3333
## Balanced Accuracy      1.0000           0.8906           0.9386

```

so the confusion matrix is

50 0 0

0 50 0

0 14 36

So the accuracy is

```
(50+50+36)/150
```

```
## [1] 0.9066667
```

Centroid

```
# fit
H.fit <- hclust(
  d,
  method = "centroid"
)
# plot
plot(H.fit)

# Add rectangles for k clusters
rect.hclust(
  H.fit,
  k=3,
  border = "red"
)
```

Cluster Dendrogram



```
groups <- cutree(
  H.fit,
  k=3
)

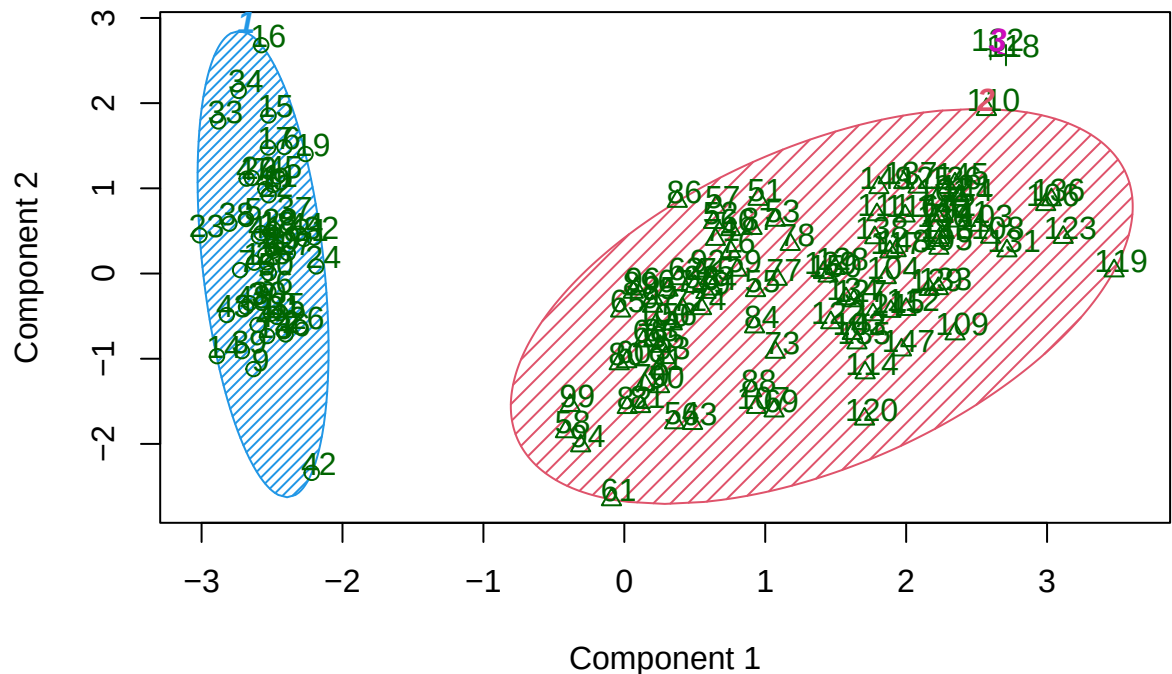
clusplot(
  data,
```

```

groups,
main = "2D representation of the Cluster solution",
color = TRUE,
shade = TRUE,
labels = 2,
lines = 0
)

```

2D representation of the Cluster solution



2D Cluster Plot

These two components explain 95.02 % of the point variability.

```

table(
  data$Species,
  groups
)

```

Confusion Matrix

```

##           groups
##           1  2  3
##  setosa    50  0  0
##  versicolor 0 50  0
##  virginica  0 48  2

```

```

clusters <- factor(
  groups,
  levels =1:3,

```

```

labels = c("setosa","versicolor","virginica")
)

confusionMatrix(
  data$Species,
  clusters
)

```

```

## Confusion Matrix and Statistics
##
##              Reference
## Prediction  setosa versicolor virginica
##   setosa      50          0          0
##   versicolor  0          50          0
##   virginica   0          48          2
##
## Overall Statistics
##
##              Accuracy : 0.68
##              95% CI : (0.599, 0.7537)
##   No Information Rate : 0.6533
##   P-Value [Acc > NIR] : 0.276
##
##              Kappa : 0.52
##
##   McNemar's Test P-Value : NA
##
## Statistics by Class:
##
##              Class: setosa Class: versicolor Class: virginica
## Sensitivity              1.0000              0.5102              1.00000
## Specificity              1.0000              1.0000              0.67568
## Pos Pred Value           1.0000              1.0000              0.04000
## Neg Pred Value           1.0000              0.5200              1.00000
## Prevalence               0.3333              0.6533              0.01333
## Detection Rate           0.3333              0.3333              0.01333
## Detection Prevalence     0.3333              0.3333              0.33333
## Balanced Accuracy        1.0000              0.7551              0.83784

```

so the confusion matrix is

50 0 0

0 50 0

0 48 2

So the accuracy is

```
(50+50+2)/150
```

```
## [1] 0.68
```

Compare K-means and Ward's

```
H.fit <- hclust(
  d,
  method="ward.D"
)

groups <- cutree(
  H.fit,
  k=3
)

ward_labels <- factor(
  groups,
  levels = 1:3,
  labels = c("setosa", "versicolor", "virginica")
)

kmeans_labels <- factor(
  kmeans_fit$cluster,
  levels = c(3,1,2),
  labels = c("setosa", "versicolor", "virginica")
)

table(
  kmeans = kmeans_labels,
  ward = ward_labels
)
```

```
##           ward
## kmeans    setosa versicolor virginica
## setosa      0         12         35
## versicolor  50          0          0
## virginica   0         52          1
```

Which is Better?

Ward's method is a little bit better than K-means, especially to find virginica level.

Performance:

- Ward > K-means > Complete > Average > Single = Centroid

I would recommend the cluster analysis as a good way to deal with this IRIS dataset.

K-means and complete methods are usually great, but here Ward method is the best one from the confusion matrix.

PCA

Compute PC's

```
iris.pca <- prcomp(  
  scaled_data,  
  center = TRUE,  
  scale.=TRUE  
)
```

```
summary(iris.pca)
```

```
## Importance of components:  
##                PC1      PC2      PC3      PC4  
## Standard deviation    1.7084 0.9560 0.38309 0.14393  
## Proportion of Variance 0.7296 0.2285 0.03669 0.00518  
## Cumulative Proportion 0.7296 0.9581 0.99482 1.00000
```

What percentage of the variations each PC would explain?

```
variance_explained <- iris.pca$sdev^2 / sum(iris.pca$sdev^2)  
variance_explained * 100
```

```
## [1] 72.9624454 22.8507618  3.6689219  0.5178709
```

PC1 -> 73%

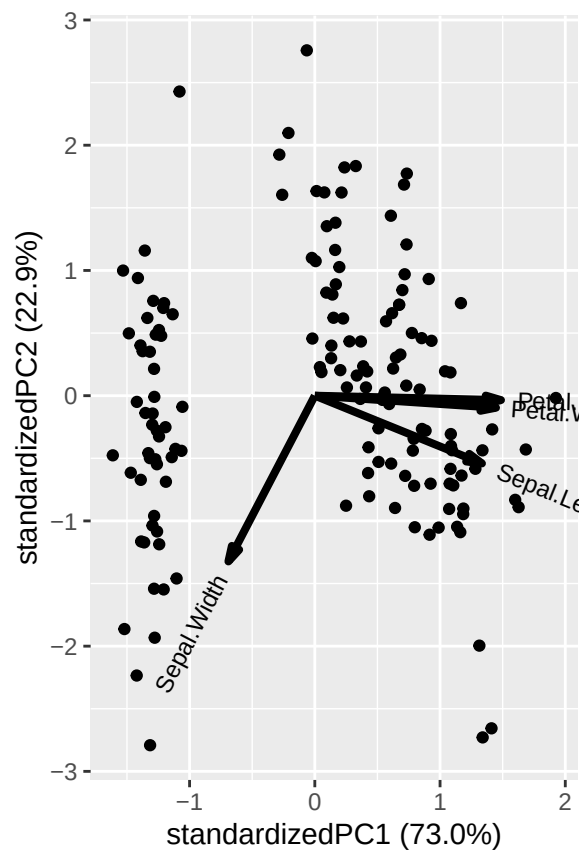
PC2 -> 23%

PC3 -> 3.5%

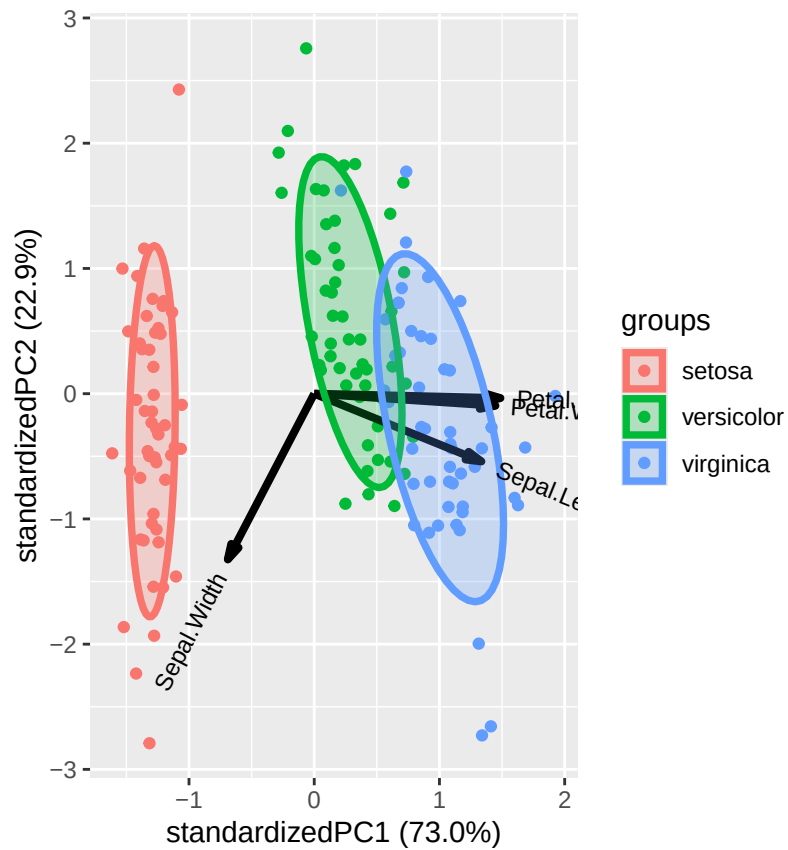
PC4 -> 0.5%

Biplot

```
ggbiplot(iris.pca)
```



```
ggbiplot(
  iris.pca,
  ellipse=TRUE,
  groups=iris$Species
)
```



Print PC1 as Linear Combination of features

```
iris.pca$rotation[,1]
```

```
## Sepal.Length Sepal.Width Petal.Length Petal.Width
## 0.5210659 -0.2693474 0.5804131 0.5648565
```

Result

I recommend the PCA method as an efficient way here. PCA is a very effective tool in this context.

The first two PCs account for over 95% of the variance in the original data, allowing us to visualize and classify the species clearly.

PCA helps simplify complex data while preserving important structure and is especially useful when dealing with high-dimensional or noisy data.